

Models of Panel Data

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Intro

Here we will study models that deal with data sets that combine time series and cross sections: **panel data sets**.

The analysis of panel data allows us to learn about the economic processes while accounting for both **heterogeneity** across units of observations and for the dynamic effects that are not visible in cross-sections.

Panel Data Modeling

panel data models are more oriented toward cross-section analyses

- they are wide but typically short.

Thus, in the typical panel, there are a large number of cross-sectional units and only a few periods.

The General Modeling Framework for Analyzing Panel Data

The basic framework is a regression model of the form

$$\begin{aligned}y_{it} &= \mathbf{x}_{it}'\beta + \mathbf{z}_i'\alpha + \varepsilon_{it} \\ &= \mathbf{x}_{it}'\beta + c_i + \varepsilon_{it}\end{aligned}$$

There are K regressors in $\mathbf{x}_{it}$, not including the constant term.

The heterogeneity, or individual effect, is $\mathbf{z}_i'\alpha$ where $\mathbf{z}_i$ contains a constant term and a set of individual or group-specific variables, which may be observed, such as race, sex, location, and so on; or unobserved, such as family specific characteristics, individual heterogeneity in skill or preferences, and so on, all of which are taken to be constant over time t .

The General Modeling Framework for Analyzing Panel Data

Notice that if all variables in $\mathbf{z}$ are observable, we could estimate the model above by OLS.

The complication, however, arises when c_i is **unobserved**.

Our main objective here will be consistent and efficient estimation of the **partial effects**

$$\beta = \partial E[y_{it} | \mathbf{x}_{it}] / \partial \mathbf{x}_{it}$$

Whether this is possible depends on the assumptions about the unobserved (latent) effects.

The General Modeling Framework for Analyzing Panel Data

We can begin with a **strict exogeneity** assumption for the independent variables,

$$E[\varepsilon_{it} | \mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, c_i] = E[\varepsilon_{it} | \mathbf{X}_i, c_i] = 0$$

This implies that the current disturbance is uncorrelated with the independent variables in every period, past, present, and future.

This assumption is sometimes unnecessarily strong but it is a convenient starting point and useful in some applications.

The General Modeling Framework for Analyzing Panel Data

A looser assumption of contemporaneous exogeneity is sometimes useful:

$$E[\varepsilon_{it} | \mathbf{x}_{it}, c_i] = 0$$

The regression model with this assumption restricts influences of $\mathbf{x}$ on $E[\mathbf{y} | \mathbf{x}, c]$ to the current period.

The General Modeling Framework for Analyzing Panel Data

The crucial aspect of the model concerns the heterogeneity.

A conventional assumption is the **mean independence**,

$$E[c_i | \mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots,] = \alpha$$

This assumption, which underlines the *random effects* model that we will discuss later, states that the unobserved variables are uncorrelated with the included variables.

This is also a strong assumption and likely to be violated in many empirical applications.

A weaker assumption would be that

$E[c_i | \mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots,] = h(\mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots,) = h(\mathbf{X}_i)$ for some unspecified but nonconstant function of $\mathbf{X}_i$.

Model Structures

Panel data models can broadly be arranged as follows:

1. **Pooled regression:** If $\mathbf{z}_i$ contains only a constant term, then the OLS estimator provides consistent and efficient estimates of the common α and slope vector β .
2. **Fixed effects:** If $\mathbf{z}_i$ is unobserved, but correlated with $\mathbf{x}_{it}$, then the OLS estimator of β is biased and inconsistent (why?). However, the model

$$y_{it} = \mathbf{x}_{it}'\beta + \alpha_i + \varepsilon_{it},$$

where $\alpha_i = \mathbf{z}_i'\alpha$ embodies all the observable effects and specifies an estimable conditional mean.

Model Structures

Panel data models can broadly be arranged as follows:

3. **Random effects:** If the observed individual heterogeneity is uncorrelated with $\mathbf{x}_{it}$, the model may be formulated as

$$\begin{aligned}y_{it} &= \mathbf{x}_{it}'\beta + E[\mathbf{z}_i'\alpha] + \mathbf{z}_i'\alpha - E[\mathbf{z}_i'\alpha] + \varepsilon_{it} \\ &= \mathbf{x}_{it}'\beta + \alpha + u_i + \varepsilon_{it}\end{aligned}$$

that is, a linear regression model with a compound disturbance that may consistently, albeit inefficiently, be estimated by LS.

Balanced and Unbalanced Panels

As suggested by the preceding discussion, a panel data set consists of n set of observations on individuals.

If each individual is observed the same number of times T , the data set is a **balanced panel**.

An **unbalanced panel** data set is one in which individuals may be observed different number of times.

- In empirical applications, this might be due to **attrition.

A **fixed panel** is one in which the same set of individuals is observed for the duration of the study.

A **rotating panel** is one in which the cast of individuals changes from one period to the next.

Well-Behaved Panel Data

Throughout the analysis we consider assumptions A.1. through A.4., as well as the asymptotic conditions on the matrices $\mathbf{X}'\mathbf{X}/n$ and $(\mathbf{X}'\varepsilon)/\mathbf{n}$.

Exceptions to the assumptions are likely to arise in a panel data set.

- For example, the $\mathbf{x}_i$'s may be correlated across observations as they might be the same individual observed in multiple periods.

The panel data set could be treated as follows:

- Since the total number of rows in $\mathbf{X}$ is now $N = n \times T$, the matrix $\bar{\mathbf{Q}}_n$ is

$$\bar{\mathbf{Q}}_n = \frac{1}{n} \sum_{i=1}^n \frac{1}{T} \mathbf{x}_i' \mathbf{x}_i = \frac{1}{n} \sum_{i=1}^n \mathbf{Q}_i$$

We then view the set of observations on the i th unit as if they were a single observation and apply our convergence arguments to the number of units increasing without bound.

The Pooled Regression Model

The simplest version of the model assumes

$$y_{it} = \alpha + \mathbf{x}_{it}'\beta + \varepsilon_{it}, i = 1, \dots, n; t = 1, \dots, T_i$$

$$E[\varepsilon_{it} | \mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{iT_i}] = 0$$

$$E[\varepsilon_{it}\varepsilon_{js} | \mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{iT_i}] = \sigma_\varepsilon^2 \text{ if } i = j \text{ and } t = s \text{ and } = 0 \text{ if } i \neq j \text{ or } t \neq s$$

Least Squares Estimation of the Pooled Model

In the form above, if the remaining assumptions of the classical model are met, then no further analysis is needed.

OLS is the efficient estimator and inference can reliably proceed along the lines of the methods developed earlier in the course.

Robust Covariance Matrix

Suppose we stack the T_i observations for individual i in a single equation

$$\mathbf{y}_i = \mathbf{x}_i\beta + \mathbf{w}_i$$

In a panel data set, although heteroskedasticity might be present, the more substantive effect is cross-observation correlation (or autocorrelation).

Let's assume that the disturbances vector consists of ε_{it} plus an omitted individual component. Then,

$$\begin{aligned} Var[\mathbf{w}_i|\mathbf{X}_i] &= \sigma_\varepsilon^2 \mathbf{I}_{T_i} + \boldsymbol{\Sigma}_i \\ &= \boldsymbol{\Omega}_i \end{aligned}$$

Robust Covariance Matrix

The OLS estimator of β is

$$\begin{aligned}\mathbf{b} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\ &= \left[\sum_{i=1}^n \mathbf{X}_i' \mathbf{X}_i \right]^{-1} \sum_{i=1}^n \mathbf{X}_i' \mathbf{w}_i\end{aligned}$$

Consistency of $\mathbf{b}$ can be established as before.

The true covariance matrix would take the form we saw for the GLS model

$$Asy.Var[\mathbf{b}] = \frac{1}{n} \text{plim} \left[\frac{1}{n} \sum_{i=1}^n \mathbf{X}_i' \mathbf{X}_i \right]^{-1} \text{plim} \left[\sum_{i=1}^n X_i' \boldsymbol{\Omega}_i X_i \right] \text{plim} \left[\frac{1}{n} \sum_{i=1}^n \mathbf{X}_i' \mathbf{X}_i \right]^{-1}$$

As before, the center matrix must be estimated.

Robust Covariance Matrix

We can estimate $Asy.Var[\mathbf{b}]$ with

$$Est.Asy.Var[\mathbf{b}] = \frac{1}{n} \text{plim} \left[\frac{1}{n} \sum_{i=1}^n \mathbf{X}_i' \mathbf{X}_i \right]^{-1} \text{plim} \left[\sum_{i=1}^n X_i' \hat{\mathbf{w}} \hat{\mathbf{w}}_i' X_i \right] \text{plim} \left[\frac{1}{n} \sum_{i=1}^n \mathbf{X}_i' \mathbf{X}_i \right]$$

where $\hat{\mathbf{w}}_i$ is the vector of T_i residuals for individual i .

This construction is analogous to the “cluster”-robust estimator developed earlier.

Bootstrapped Covariance Matrix

Bootstrapping offers another approach to estimating an appropriate covariance matrix for the estimator.

Suppose we have n individuals (groups) over T periods.

To compute the **block bootstrap** estimator, we draw random samples of $N = n$ with *repetition*. Then, we repeat the process R times.

After R repetitions, we compute the empirical variance of the R replicates.

The estimator is

$$Est.Asy.Var[\mathbf{b}] = \frac{1}{R} \sum_{r=1}^R (\mathbf{b}_r - \bar{\mathbf{b}})(\mathbf{b}_r - \bar{\mathbf{b}})'$$

Robust Estimation Using Group Means

The pooled regression model can also be estimated using the sample means of the data.

Pre-multiplying the regression model by $(1/T)\mathbf{i}'$, where $\mathbf{i}'$ is a row vector of ones,

$$(1/T)\mathbf{i}'\mathbf{y}_i = (1/T)\mathbf{i}'\mathbf{x}_i\beta + (1/T)\mathbf{i}'\mathbf{w}_i$$

or

$$\bar{y}_i = \bar{\mathbf{x}}_i'\beta + \bar{w}_i$$

Heteroskedastic variances are now $\sigma_i^2 = (1/T^2)\mathbf{i}'\mathbf{\Omega}_i\mathbf{i}$.

Why might one want to use this estimator when the full data set is available?

Robust Estimation Using Group Means

Consider the general model

$$\mathbf{y}_i = \mathbf{x}_i\beta + c_i\mathbf{i} + \mathbf{w}_i,$$

where c_i is the (latent) heterogeneous effect.

If the mean independence assumption, $E[c_i|\mathbf{X}_i] = \alpha$ is not met, then the effect will be transmitted to the group means as well.

Robust Estimation Using Group Means

A common specification is Mundlak's (1978), where we employ the projection of c_i on the group means

$$c_i | \mathbf{X}_i = \bar{\mathbf{x}}_i' \gamma + \nu_i$$

Then,

$$\begin{aligned} y_{it} &= \mathbf{x}_{it}' \beta + c_i + \varepsilon_{it} \\ &= \mathbf{x}_{it}' \beta + \bar{\mathbf{x}}_i' \gamma + [\nu_i + \varepsilon_{it}] \\ &= \mathbf{x}_{it}' \beta + \bar{\mathbf{x}}_i' \gamma + u_{it}, \end{aligned}$$

where, by construction, $Cov[u_{it}, \bar{\mathbf{x}}_i] = 0$.

The implication is that the group means estimator estimates not β , but $\beta + \gamma$.

Averaging the observations in the group collects the entire set of effects, observed and latent, in the group means.

Estimator with First Differences

First differencing is another approach to estimation.

Consider the model

$$y_{it} = c_i + \mathbf{x}_{it}'\beta + \varepsilon_{it}$$

which implies the first differences equation

$$\Delta y_{it} = \Delta c_i + (\Delta \mathbf{x}_{it})'\beta + \Delta \varepsilon_{it}$$

or

$$\begin{aligned}\Delta y_{it} &= (\Delta \mathbf{x}_{it})'\beta + \varepsilon_{it} - \varepsilon_{i,t-1} \\ &= (\Delta \mathbf{x}_{it})'\beta + u_{it}\end{aligned}$$

Estimator with First Differences

The advantage of the first difference approach is that it removes the heterogeneity from the model.

The disadvantage is that it removes all time-invariant variables from the model.

- If we have a panel data set and want to estimate the effects of schooling on wages, for example, we would not be able to employ this strategy, as years of education of workers are usually fixed in time.

Note as well that the differencing procedure trades the cross-observation correlation in c_i for a moving average (MA) disturbances $u_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}$, which is autocorrelated.

However, first differencing might be useful when we have a setting involving two time periods, before and after a treatment.

Estimating Treatment Effects with First Differencing

Consider the model

$$y_{it} = c_i + \mathbf{x}'_{it}\beta + \theta S_{it} + \varepsilon_{it},$$

where $t = 1, 2$, and $S_{it} = 0$ is period 1 and $S_{it} = 1$ in period 2.

The treatment effect would be $E[\Delta y_i | \Delta \mathbf{x}_i = 0]$. So, taking the first difference, we find

$$\Delta y_i = \theta + (\Delta \mathbf{x}_{it})' \beta + u_i$$

For $\Delta \mathbf{x}_i = 0$, the treatment effect is θ .

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